

NETWORK ANALYSIS

To Plan, monitor and control process/measures for projects

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Project

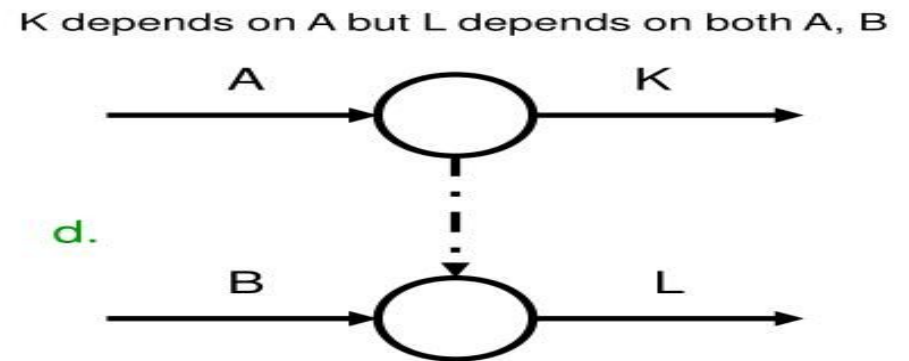
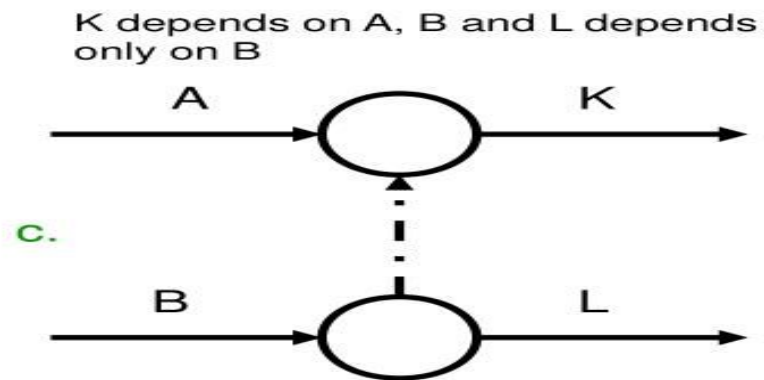
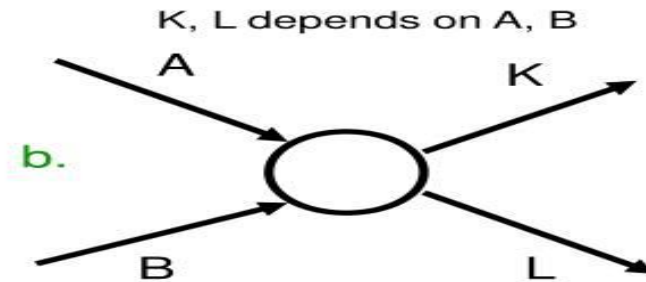
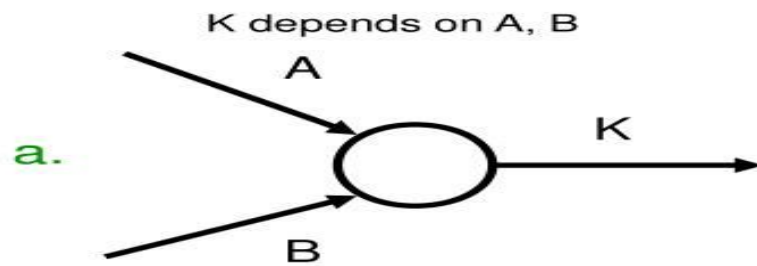
- Project- defined as a unique activity which has a definite beginning and definite end, consumes time and resources.
- Project- is a set of inter related activities organized for a common goal.
- Project is any human undertaking with a clear beginning and clear ending.
- Examples: ?

Network

- Interrelationship of activities or events of a project with a mention on time required for the completion of each activity.
- Is a schematic representation of work to be done (activities) drawn in such a way that relationships between work items are logically defined.

Activity

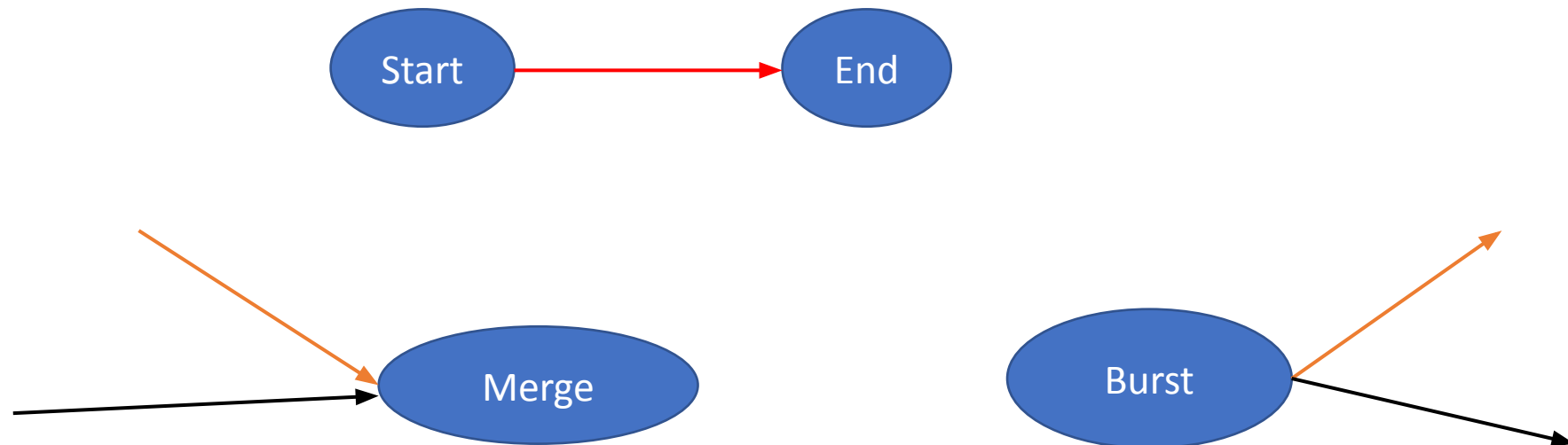
Activity on Arrow (AoA) Network Diagrams



Activity

- Predecessor
- Successor
- Critical
- Dummy

Events



Dummy activity

10. Parallel activities between two events, without intervening events, are prohibited. When two or more parallel activities in a project have the same head and tail events, dummy activities are needed in constructing the network.

Dummy activities do not consume time or resources. An efficient network contains a minimum number of dummy activities required to portray the correct precedence relationships.

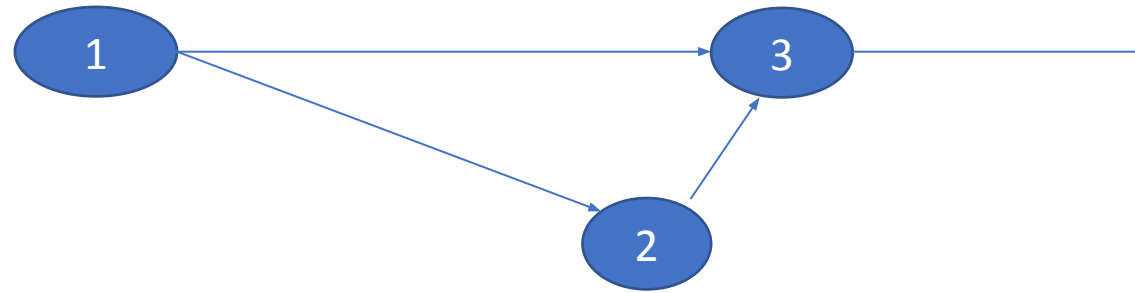
11. Looping is not permitted in a network.

Dummy activity

- Wrong



- Correct



Events

- Start event
- End event
- Merge event
- Burst event

Project network -Methods

- CPM- Critical Path Method
- PERT-Programme Evaluation and Review Technique

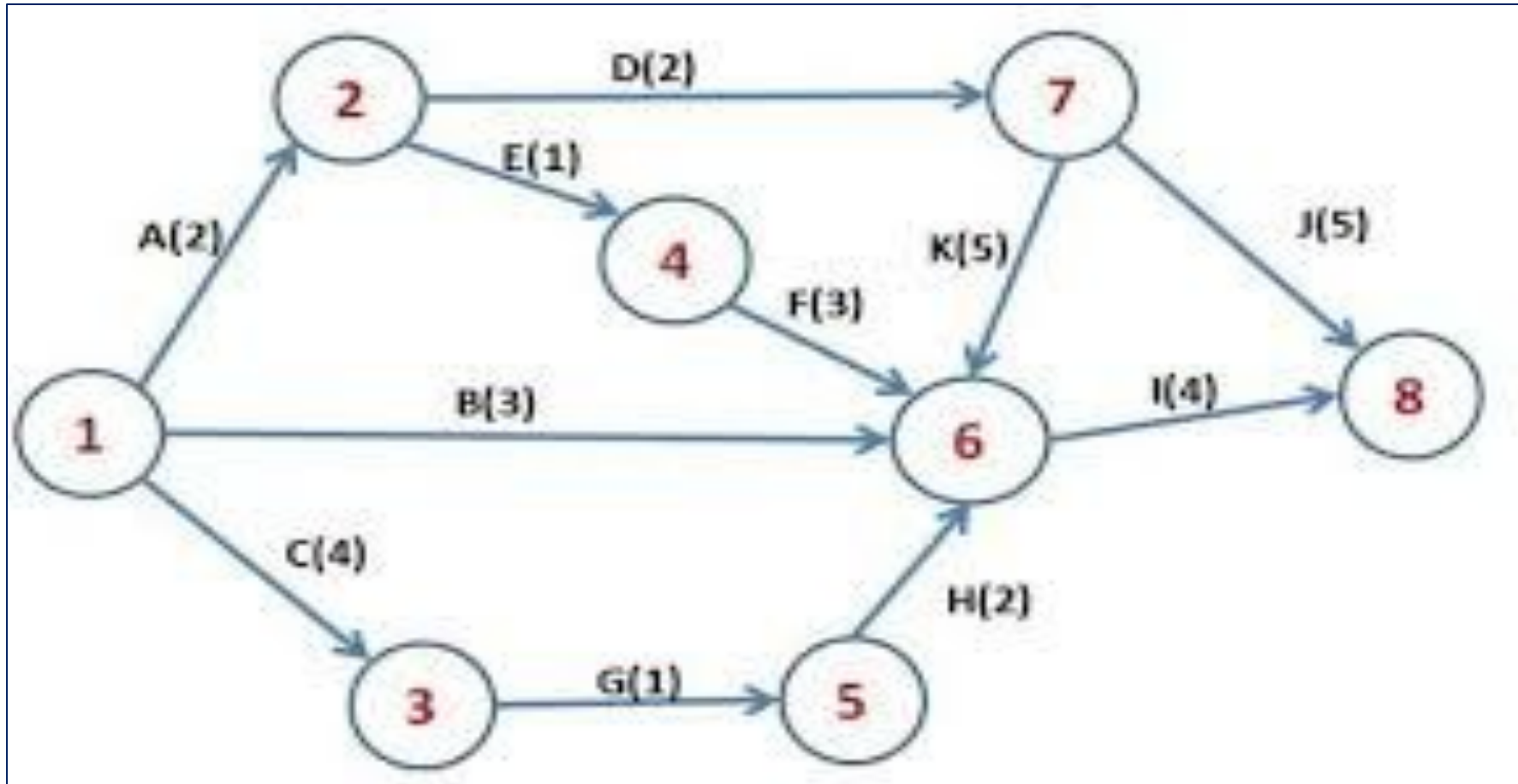
CPM

- Activity oriented
- Deterministic model
- Single time estimate possible
- Best suited for construction

PERT

- Event oriented
- Probabilistic model
- 3 time estimate (T_p , T_o and T_m)
- Expected time (T_e) = $(T_p + 4T_m + T_o)/6$
- Research and development projects

Project Network



Network paths

Path (activity based)B-I	Path (event based)	Duration
A-D-J	1-2-7-8	9
A-E-F-I	1-2-4-6-8	10
B-I	1-6-8	7
C-G-H-I	1-3-5-6-8	11
A_D_K_I	1-2-7-6-8	13

Critical path

- Path having maximum duration
- This duration is the minimum time required for the completion of project.
- Activities in the critical path are the critical activities.
- This path is represented by a double line.

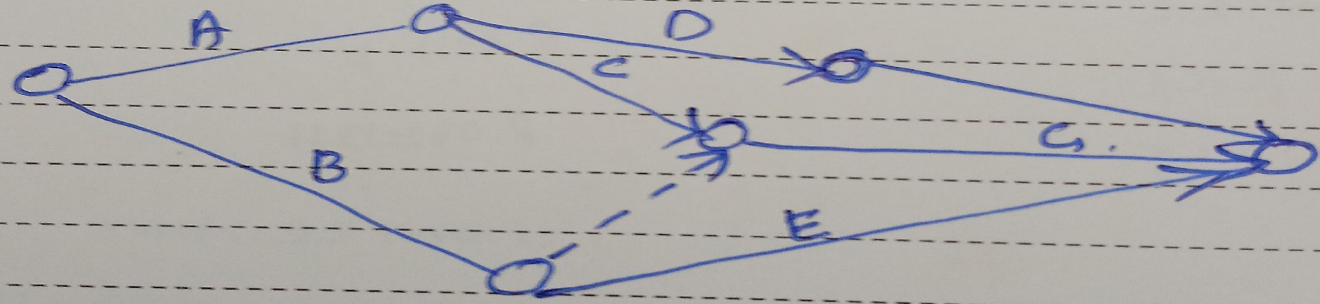
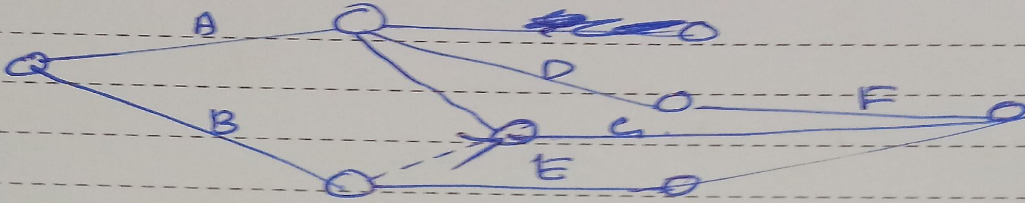
problem

ACTIVITY	IMMEDIATE PREDECESSOR	ACTIVITY DURATION (DAYS)
A	-	3
B	A	5
C	A	4
D	B	3
E	C	5
F	C	2
G	D,E,F	6
H	G	1

CPM-NETWORK-2

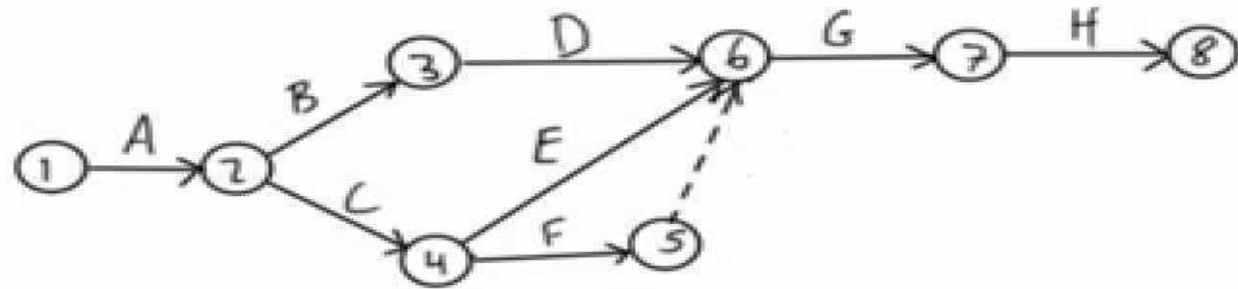
ACTIVITY	IMMEDIATE PREDECESSOR	ACTIVITY DURATION (DAYS)
A	-	5
B	-	4
C	A	3
D	A	6
E	B	3
F	D	4
G	B,C	4

P

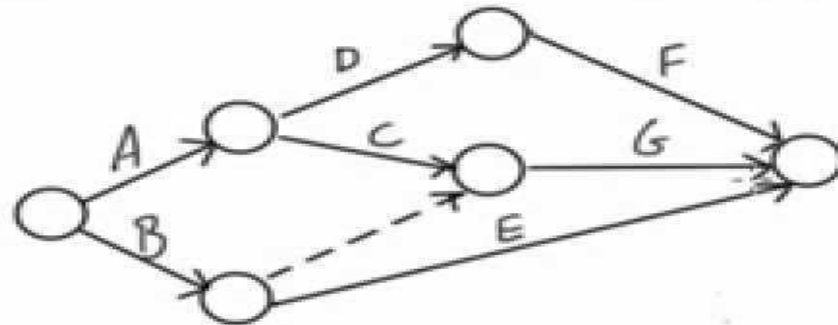


Network

Activity	Predecessor
A	-
B	A
C	A
D	B
E	C
F	C
G	D,E,F
H	G



Activity	Predecessor
A	-
B	-
C	A
D	A
E	B
F	D
G	B,C



CRASHING (TIME COST TRADE OFF)

- The process of reducing project duration is called crashing
- Crashing will be done at the cost of extra resources....in critical path.
- Any reduction in project time means reduction in indirect costs.
- Direct costs?
- Indirect costs?

Direct and Indirect Costs

- Direct costs: material cost, machinery used in the project, labour-both direct and subcontracting.
- Indirect costs: Overheads, office and administrative expenses, penalty cost. This varies with time

Cost slope

REDMI 9 PRIME
AI QUAD CAMERA

1.65 = 3

20.8. TIME-COST TRADE OFF (CRASHING)
The total overall cost will consist of direct and indirect costs. The indirect cost consists of overheads, office expenditures, administrative expenses, cost penalty for delay in work, etc., the indirect time varies with time. Any reduction in project time means reduction in indirect costs. Direct costs include cost of materials, machinery used in the project and payments toward labour and subcontracting.
It is assumed that for each activity, there is an activity duration for which the direct cost is minimum. If activity results in more than this time, more resources and, hence, more funds are required. For instance a point will be reached beyond which no further reduction in time will be possible irrespective of the activity at minimum cost is called normal time and the minimum time for the activity is called crash time. The costs associated with these times are called normal and crash cost. A linear relationship is assumed between time and cost as shown in Fig. 20.11.

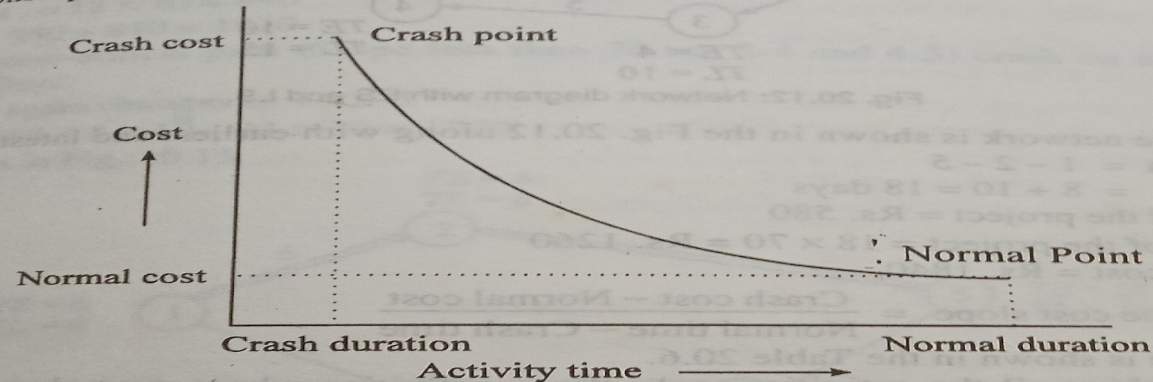


Fig. 20.11: Time-cost relationship.

$$\text{Cost slope (CS)} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal time} - \text{Crash time}}$$

Procedure for crashing —

The process of reducing the project duration is called crashing. The crashing is at the extra resources, i.e., at an extra cost.

The procedure for project crashing are:

1. Construct the network diagram and for a given network diagram find the critical
2. Calculate the cost slope for different activities.

Cost slope

- C_t - is the minimum time possible for the activity
- N_t - Time for the activity at minimum cost
- Costs associated with these times are called Crash cost (C_c) and Normal cost (N_c)
- Cost slope = $(C_c - N_c) / (N_t - C_t)$

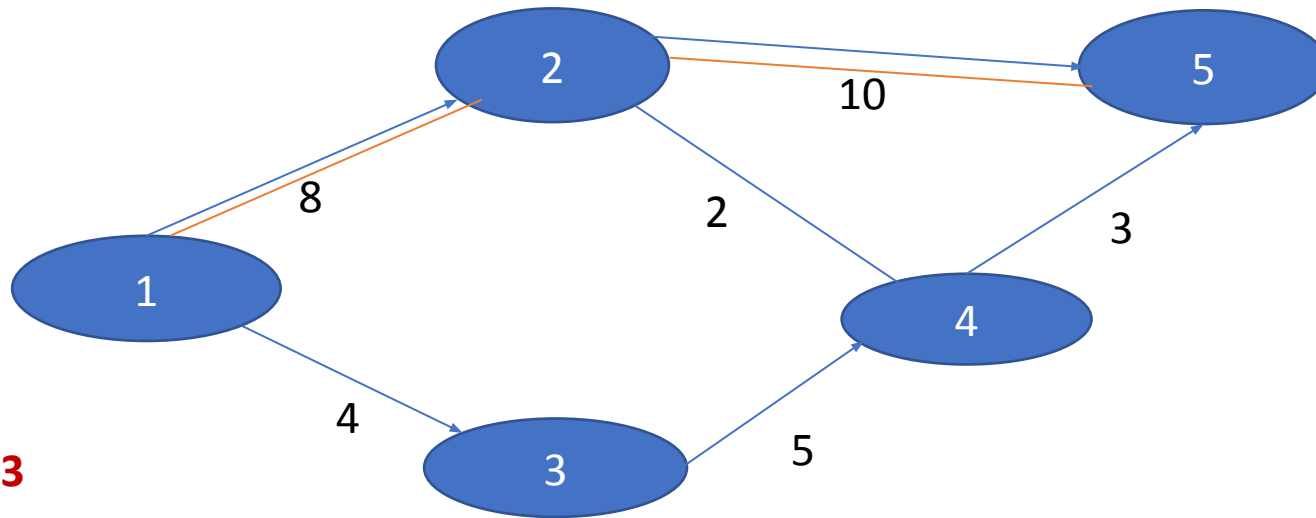
Crashing procedure

1. Construct the network diagram and identify the critical path
2. Calculate the cost slope for different activities
3. Crashing the activities in critical path as per ranking(first crash the activity with lowest cost slope)
4. Determine the total cost of the project after every crashing stages.
5. This includes both direct and indirect costs.
6. Finally find the total cost and duration after crashing all possible activities in the critical path.

Determine the optimal project duration.
Assume - indirect cost =Rs.70/hr

Normal			crash		Cost slope
Activity	Time(Hrs)	Cost(Rs.)	Time(Hrs)	Cost(Rs.)	
1-2	8	100	6	200	50
1-3	4	150	2	350	100
2-4	2	50	1	90	40
2-5	10	100	5	400	60
3-4	5	100	1	200	25
4-5	3	80	1	100	10
	Normal cost (direct cost)	580			

Crashing problem



1-2-5=18

1-2-4-5=13

1-3-4-5=12

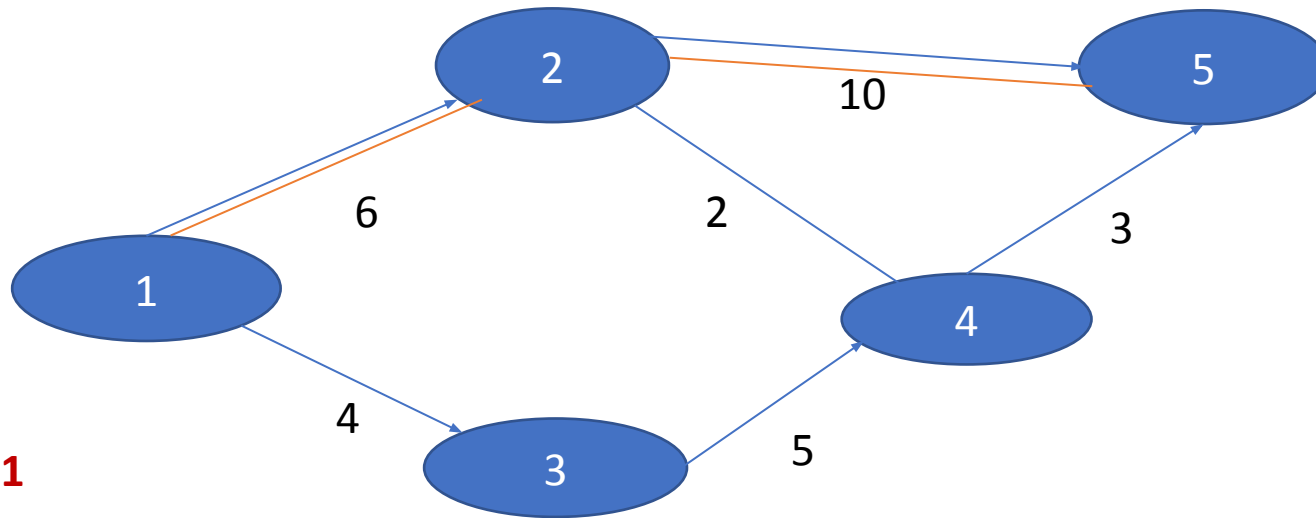
Before Crashing

- Direct cost of the project= Rs.580
- Indirect cost of the project= $18 \times 70 = 1260$
- Total cost of the project= $580 + 1260 = 1840$

Crashing 1-2

- Direct cost= $580+2\times 50=680$
- Indirect cost= $70\times 16=1120$
- Total cost=1800

Crashing 1-2



1-2-5=16

1-2-4-5=11

1-3-4-5=12

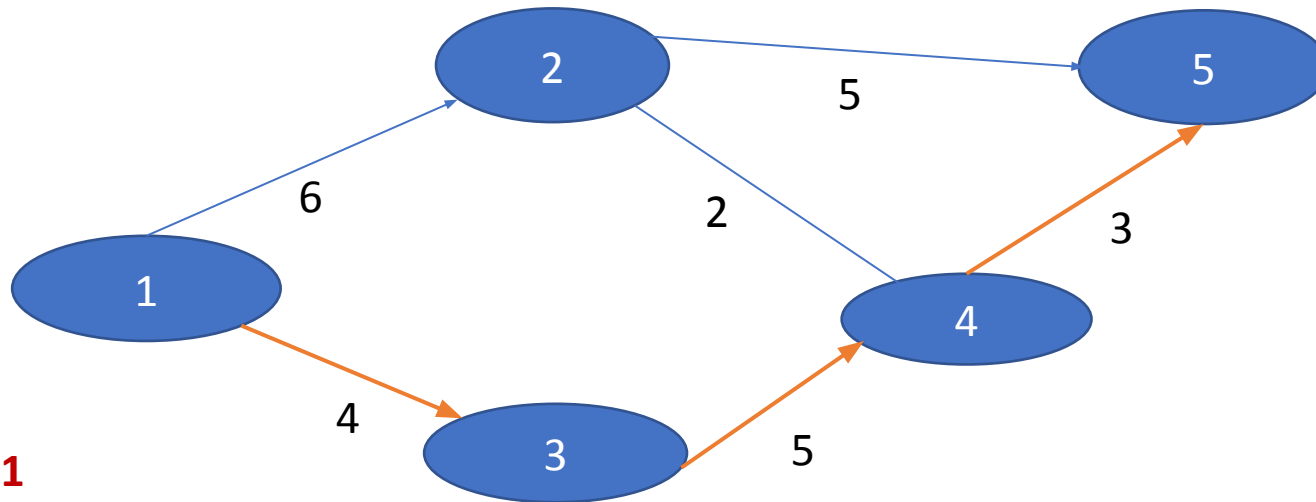
Crashing 2-5

Direct cost= $580+2\times 50+ 5\times 60=980$

Indirect cost= $70\times 12=840$

Total cost=1820

Crashing problem

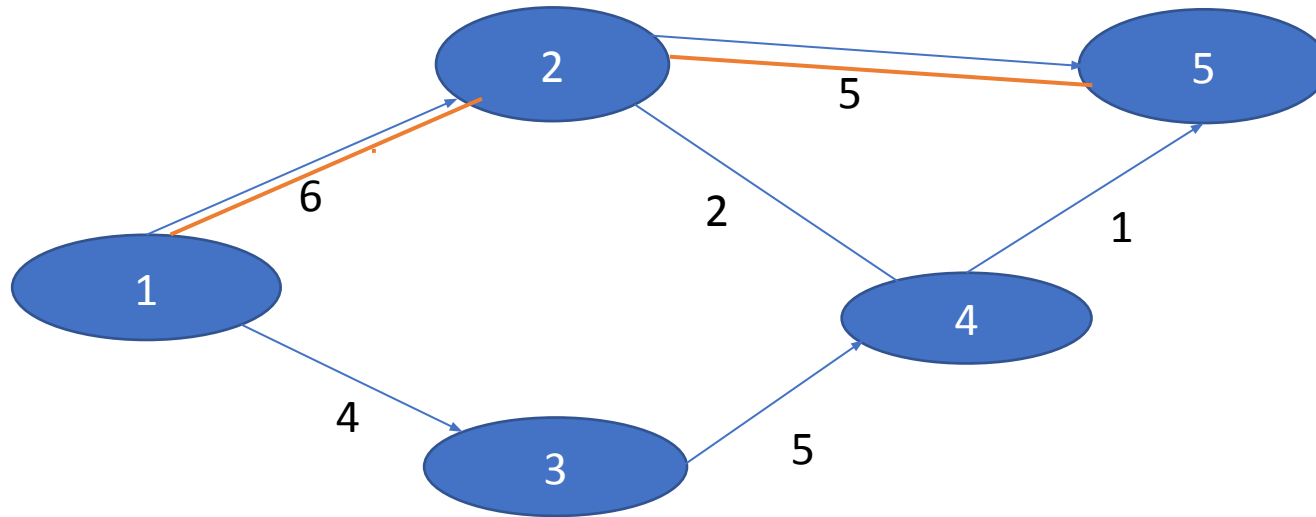


1-2-5=11

1-2-4-5=11

1-3-4-5=12

Crashing 4-5



1-2-5=11

1-2-4-5=9

1-3-4-5=10

Crashing 4-5

- Direct cost= $980 + 2 \times 10 = 1000$
- Indirect cost= $11 \times 70 = 770$
- Total project cost=1770

1760

Crashing solution

- Project can not be crashed further as the activities in the critical paths are already crashed.
- So the optimum solution= ???

PERT

- Event oriented
- Probabilistic model
- 3 time estimate (t_p , t_o and t_m)
- Expected time (t_e) = $(t_p + 4t_m + t_o)/6$
- Variance = $[(t_p - t_o)/6]^2$
- Research and development projects

PERT

Q 2. A project has the following characteristics

Activity	Most optimistic time (a)	Most pessimistic time (b)	Most likely time (m)
(1 – 2)	1	5	1.5
(2 – 3)	1	3	2
(2 – 4)	1	5	3
(3 – 5)	3	5	4
(4 – 5)	2	4	3
(4 – 6)	3	7	5
(5 – 7)	4	6	5
(6 – 7)	6	8	7
(7 – 8)	2	6	4
(7 – 9)	5	8	6
(8 – 10)	1	3	2
(9 – 10)	3	7	5

Construct a PERT network. Find the critical path and variance for each event.

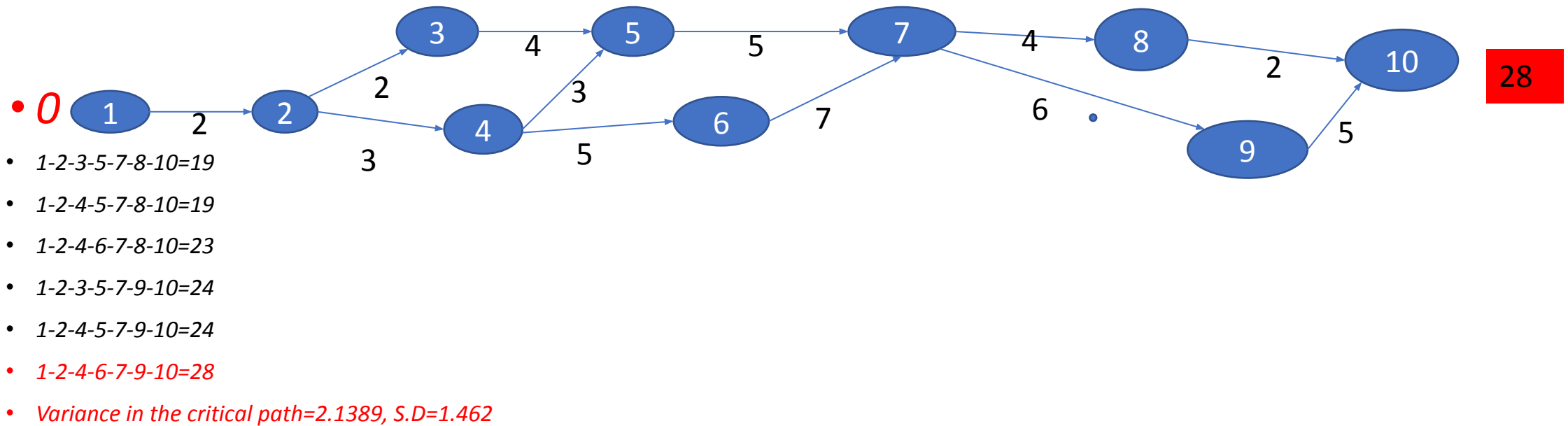
solution

Expected time(T_e)= $t_p+t_o+4t_m/6$	Variance = $[(t_p-t_o)/6]^2$	S.D=

Solution

Activity	Te	Variance	Critical path variance
1-2	2	4/9	4/9
2-3	2	1/9	
2-4	3	4/9	4/9
3-5	4	1/9	
4-5	3	1/9	
4-6	5	4/9	4/9
5-7	5	1/9	
6-7	7	1/9	1/9
7-8	4	4/9	
7-9	6	1/4	1/4
8-10	2	1/9	
9-10	5	4/9	4/9

PERT-NETWORK- Solution



Probability of completion

- Find the duration in the critical path
- Find the total variance in the critical path
- Square root of total variance will give you the standard deviation
- Compute standard normal variate $Z = (x - \mu) / \sigma$
- X is the anticipated /expected project duration under question
- μ is the duration in the critical path..mean time of project calculated
- σ is the standard deviation in the critical path.

Slack and float

- **slack** is referring to the amount of time that an activity can start later than originally planned
- **float** is about the time when an activity takes longer than originally planned. ...
- Total **float** is the amount of time that an activity can be delayed without delaying the completion of the project.

Slack Versus Float

- The terms "slack" and "float" are often used interchangeably.
- However, the main difference between float and slack is that slack is typically associated with inactivity (PERT), while float is associated with activity (CPM).
- Slack time allows an activity to start later than originally planned, while float time allows an activity to take longer than originally planned.

Slack or Float

- = Latest Start Time - Earliest Start Time

OR

Latest Finish Time - Earliest Finish Time